BirdCLEF 2026: Embeddings and Fine-Tuning

Progress report 2: from the Pantanal dataset to a fine-tuned BEATs classifier

3 October 2026 · follows *Progress report 1* (20 North American species, docs/progress_report.pdf)

Summary

- Downloaded the full **BirdCLEF+ 2026** dataset (16 GB): 35,549 recordings of **234 classes** from the Pantanal, Brazil — 162 birds plus frogs, insects, mammals and a reptile.
- Adapted the pipeline to it and produced **137,342 five-second clips** from 33,012 recordings, with train / val / test split by recording so no recording leaks across.
- Off-the-shelf (frozen) BEATs already separates the classes at **30× chance**. A classifier on those frozen embeddings reaches **55.4%** clip accuracy.
- **Fine-tuning BEATs end to end raised held-out accuracy to 72.3%** (+16.9 points), macro-F1 to 68.4% and per-recording accuracy to **81.2%**.
- Main limits: 28 classes have no training clips, the data is 98% birds, and Xeno-canto vs iNaturalist recordings still cluster by source.

Note: **BirdCLEF species barely overlap the first project**. Only House Sparrow is in both, so this is a new task, not more data for the 20-species work. The goal here is classification accuracy on our own held-out split, not a Kaggle submission.

1. The data

Item	Size	Contents
train_audio/	11 GB	35,549 recordings in 206 class folders; all match train.csv
train_soundscapes/	5.1 GB	10,658 long field recordings; only 66 are labelled (1,478 time-stamped labels)
taxonomy.csv	234 classes	162 Aves, 35 Amphibia, 28 Insecta, 8 Mammalia, 1 Reptilia
train.csv	35,549 rows	23,043 from Xeno-canto, 12,506 from iNaturalist; label, secondary labels, location, recordist, licence

Downloaded straight to a Vast.ai GPU instance (RTX 5090, 32 GB VRAM, 96 cores) by streaming the zip from Kaggle and unpacking on the fly, so the 16 GB archive never had to sit on its 32 GB disk. It took 3 minutes.

28 classes (25 insects, 3 frogs) have no training clips at all; they appear only in the labelled field recordings, so nothing below can predict them yet.

2. Preprocessing and embedding

Same audio contract as project 1 (16 kHz mono, 5 s clips, silence trim, up to 8 clips per recording, 70/15/15 split by recording), with three changes for a multi-taxon set:

- **No frequency-band filter.** It was tuned for birdsong and would discard frog and insect calls.
- **Short recordings padded, not dropped.** Recordings of 1–5 s are padded to 5 s; otherwise 7% of recordings, mostly frogs and insects, would yield nothing.
- **A `taxon_class` column** (bird / frog / insect / mammal) carried through to the Projector, plus the source (Xeno-canto vs iNaturalist).

Stage	Result
Clips	137,342 from 33,012 recordings: train 96,551 / val 20,339 / test (viz) 20,452
Skipped	2,537 recordings: 2,157 below the loudness floor, 380 under 1 s. Every class with audio kept at least one recording; owls and hawks lost the most (15–20%)
Balance	Birds 134,450 clips; frogs 1,756; insects 814 (3 classes); mammals 321; reptile 1. Clips per class: 1 to 2,168, median 496
Embeddings	BEATs iter3_plus_AS2M, frozen, 768-d: 137,342 vectors in 6.2 min on the GPU

3. Frozen embeddings and the baseline

Neighbour purity: 26.0% of each clip's 10 nearest neighbours (excluding its own recording) are the same class, against 0.9% by chance — **30× chance**. Clearest classes: Rufous Nightjar 74%, Giant Cicada 67%, White-coated Titi 65%, Rococo Toad 65%. Twelve classes score 0%, mostly frogs and mammals with 1–3 recordings.

Source bias: iNaturalist clips have 58.8% iNaturalist neighbours although they are only 24% of the data, about 2.5× more than expected. Part of this is the class mix (all insects and most frogs come from iNaturalist), but 30k of those clips are birds, so phone recordings genuinely sound different from Xeno-canto field recordings.

Baseline classifiers trained on the frozen embeddings; best epoch chosen on val, reported on the held-out viz split:

Classifier on frozen embeddings	Clip accuracy	Macro-F1	Recording accuracy
Linear	52.8%	48.8%	59.6%
Small MLP (baseline)	**55.4%**	**51.9%**	**62.8%**
Linear, class-weighted	40.1%	48.8%	40.3%
MLP, class-weighted	24.4%	41.7%	21.5%

Full inverse-frequency class weighting hurt: classes with a single clip received weights hundreds of times larger and dominated training.

4. Fine-tuning BEATs

The whole network was trained end to end with a classification head on the **train** split, the best epoch was picked by val macro-F1, and it was reported once on the held-out **viz** split — the same clips and metrics as the baseline.

Setting	Value	Why
Learning rate	5e-05 (top layer), ×0.75 per layer down; head 0.001	deeper, more general layers change less
Schedule	10 epochs, batch 128, 500 warm-up steps, cosine decay, AdamW	
Imbalance	sample each clip with weight count$^{-0.5}$	middle ground; full weighting hurt the baseline
Augmentation	random time shift, gain 0.3–1.2×, mixup (p = 0.5), label smoothing 0.1	more variety from the same clips
Hardware	RTX 5090, bf16: ~288 clips/s, 27.5 GB, 6.2 min per epoch, 62 min total	

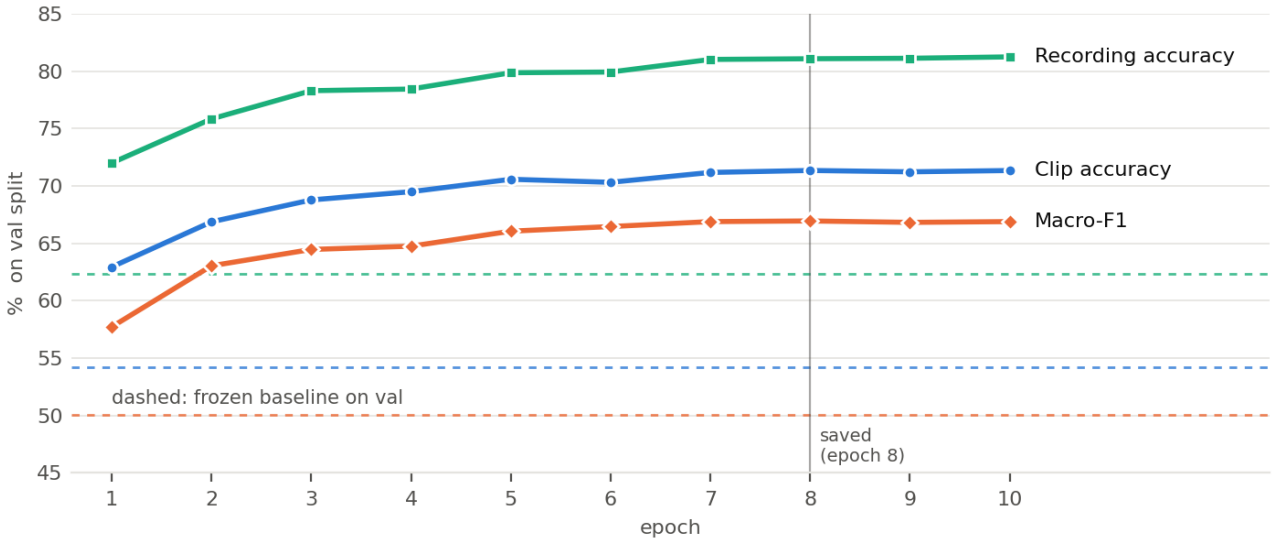


Figure 1. Validation scores per epoch. Most of the gain comes in the first 3 epochs; it levels off after epoch 7. Epoch 8 was saved.

Results on the held-out test split

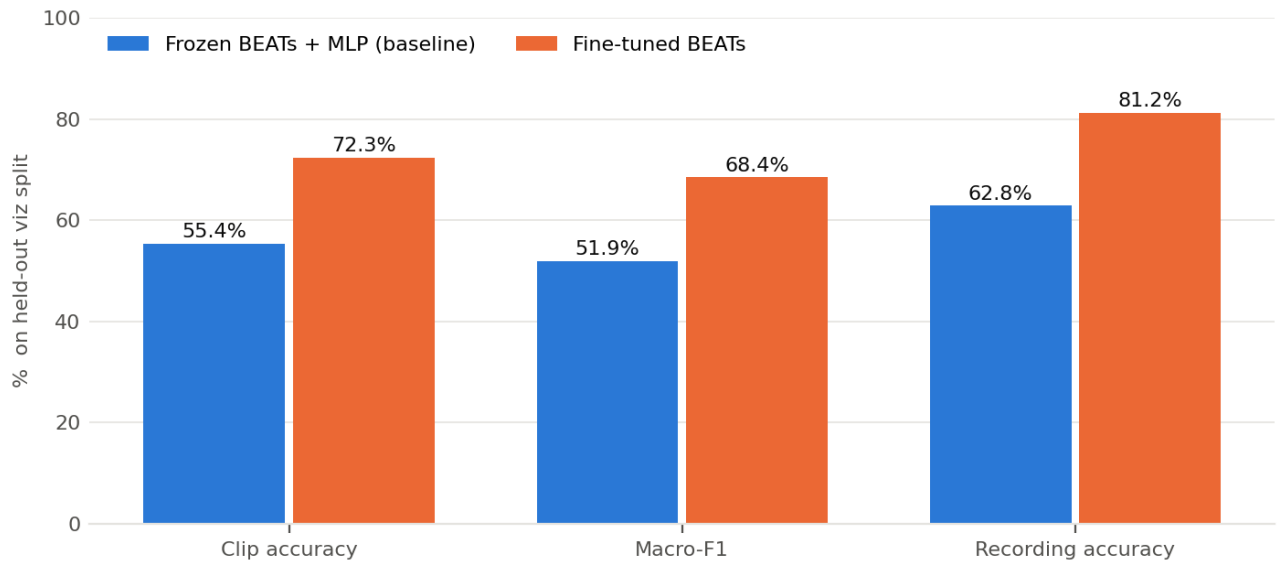


Figure 2. Frozen baseline vs fine-tuned BEATs on the viz split — recordings never seen in training.

Model (tested on viz)	Clip accuracy	Macro-F1	Recording accuracy
Frozen BEATs + MLP	55.4%	51.9%	62.8%
Fine-tuned BEATs	**72.3%**	**68.4%**	**81.2%**
Gain	+16.9 pts	+16.6 pts	+18.4 pts

Test scores sit slightly above validation, so the model is not over-tuned to val.

What changed in the embeddings

Neighbour purity	Frozen	Fine-tuned
All classes	26.0%	70.6%
Aves	23.5%	67.1%
Amphibia	18.8%	47.0%
Insecta	41.9%	83.8%
Mammalia	26.0%	41.0%

Caveat: purity is measured over all clips, including the training clips whose labels the model learned, so the fine-tuned figure is inflated. Use the test accuracy above as the honest number. The source bias did not change (81% same-source neighbours).

5. What exists now

File (on the laptop)	What it is
`data/checkpoints/bc26_ft/best.pt`	fine-tuned model, 362 MB, with its 206 class names
`data/embeddings/embeddings_bc26.npy`	frozen embeddings, 137,342 × 768
`data/embeddings/embeddings_bc26_ft.npy`	fine-tuned embeddings, 137,342 × 768
`exports/bc26_viz/`	Projector files, frozen, 20,452 test clips
`exports/bc26_ft_viz/`	Projector files, fine-tuned — load next to the frozen ones
`exports/bc26*/`	analysis.md, probe_*.json, finetune_result.json (every epoch)
`src/finetune.py, train_probe.py`	new training and baseline scripts (uncommitted)

6. Next steps for higher accuracy

- **Random 5 s crops from full recordings** instead of fixed clips: uses the audio the 8-clip cap discarded and gives fresh crops every epoch.
- **Test-time augmentation:** average predictions over several shifted crops. Cheap; usually a point or two.

- **Background noise** from the 10,592 unlabelled field recordings, for robustness to real field conditions.
- **Ensemble** of 2–3 models trained with different seeds.
- **Pseudo-labelling** the field recordings to cover the 28 classes with no training clips.

Housekeeping: the code changes are not yet committed, and the GPU instance is still running (its storage is not persistent; everything important is copied to the laptop).